Christine Zou
Project Portfolio

For website see https://christine-zou.github.io

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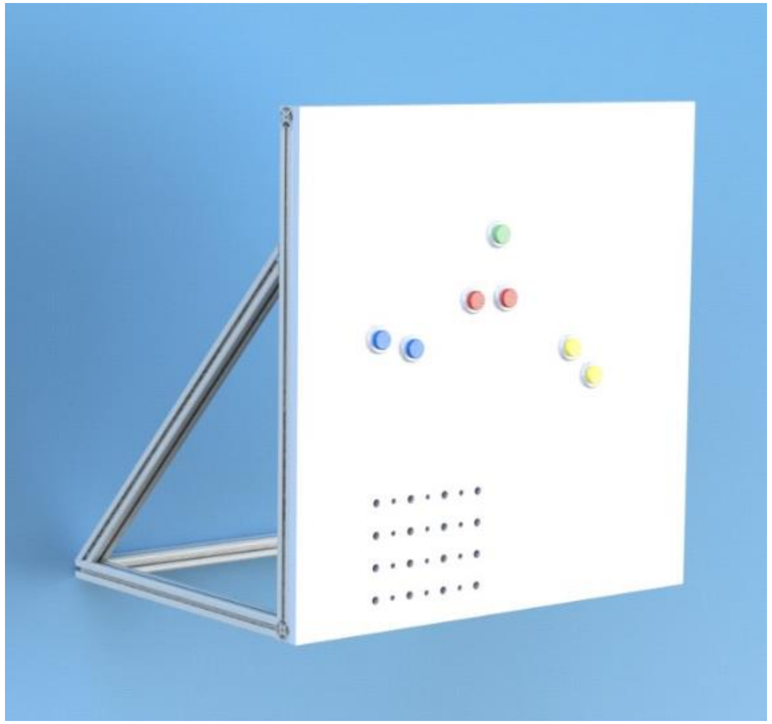
Automated Robotic Linkage Mechanism

Goal: design, build, and test a powered mechanism that will automatically press paired arcade buttons (blue, red and yellow) on a playing field as rapidly as possible in 1 minute.

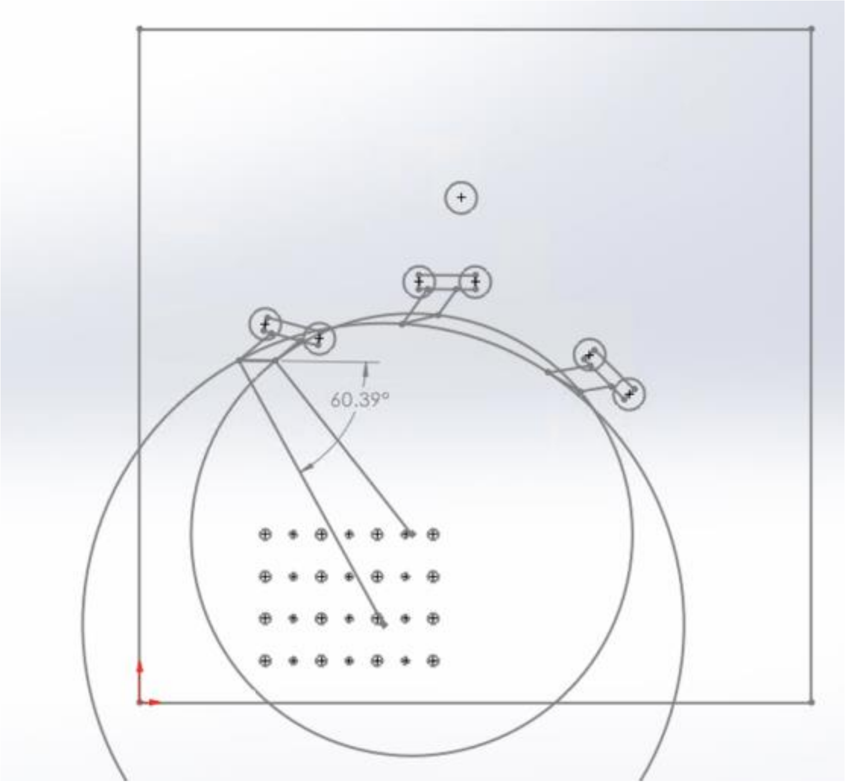


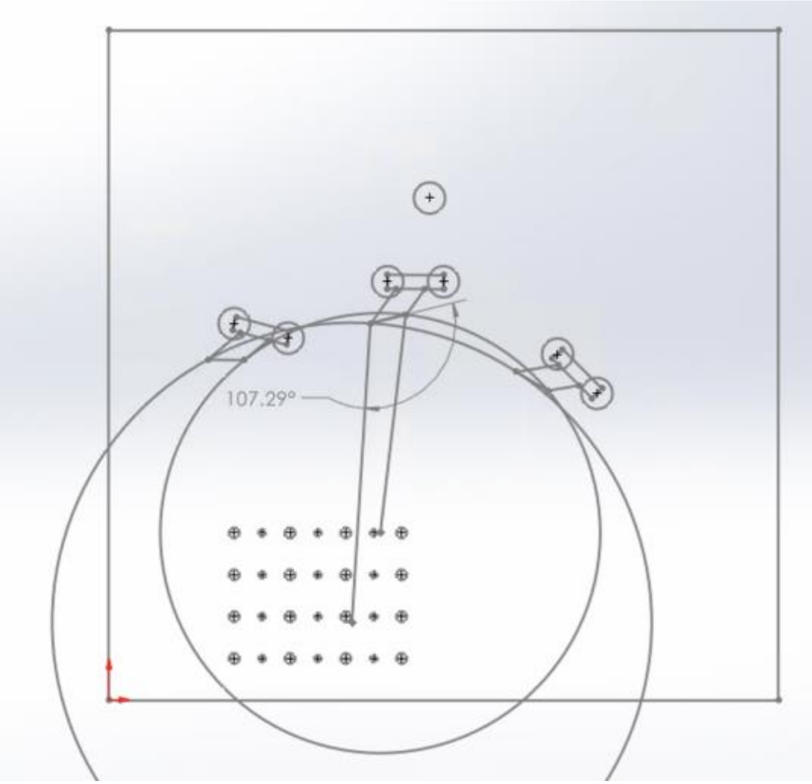
Playing field including bonus green button

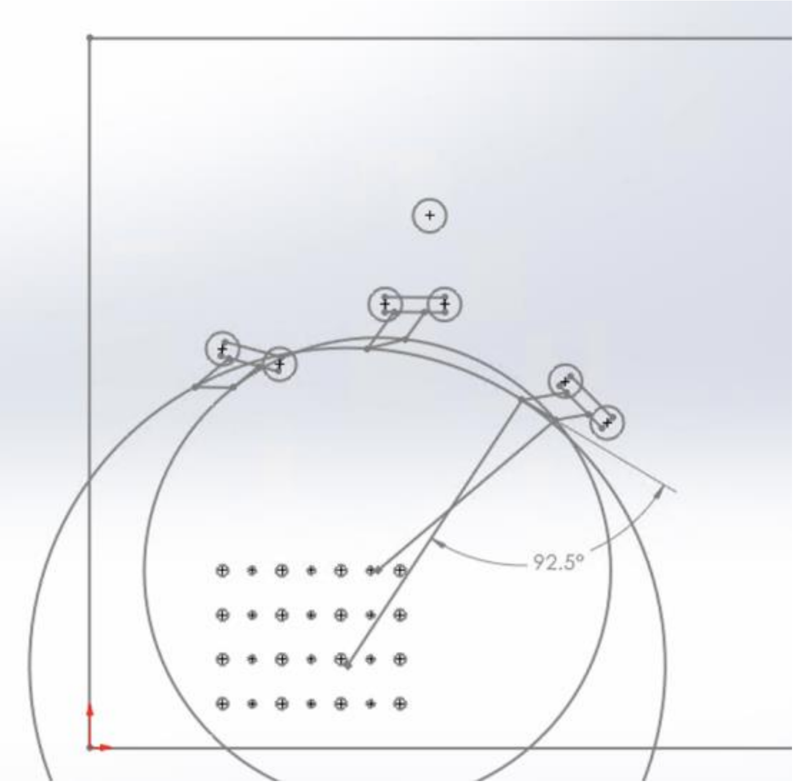
Primary Role: Integration & Testing (Electrical setup, Arduino programming, optimization, etc.), Chief Morale Officer (see slides 6-7 for details)

Skills: Motion Generation, Motor Control, Machining, Transmission Design, Arduino, Troubleshooting, CAD (SolidWorks)

Transmission Angle Design

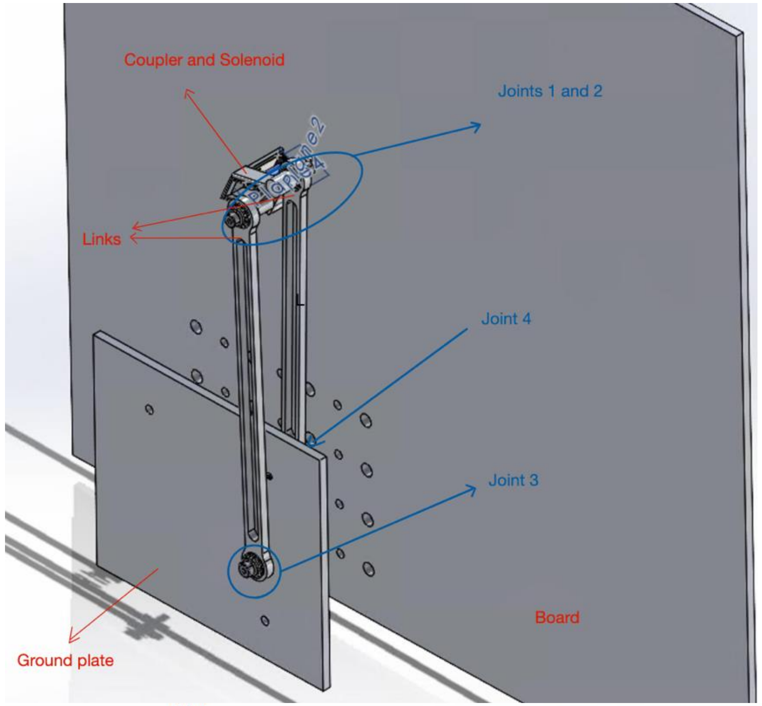


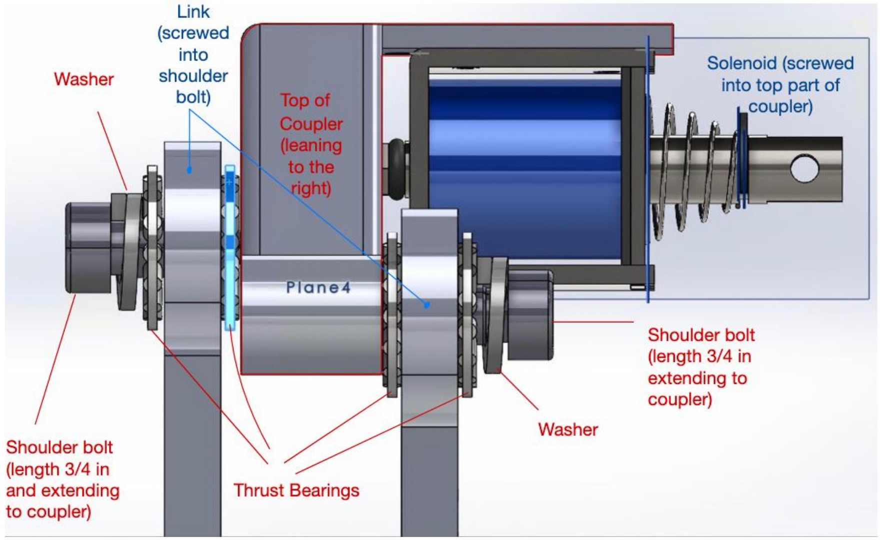


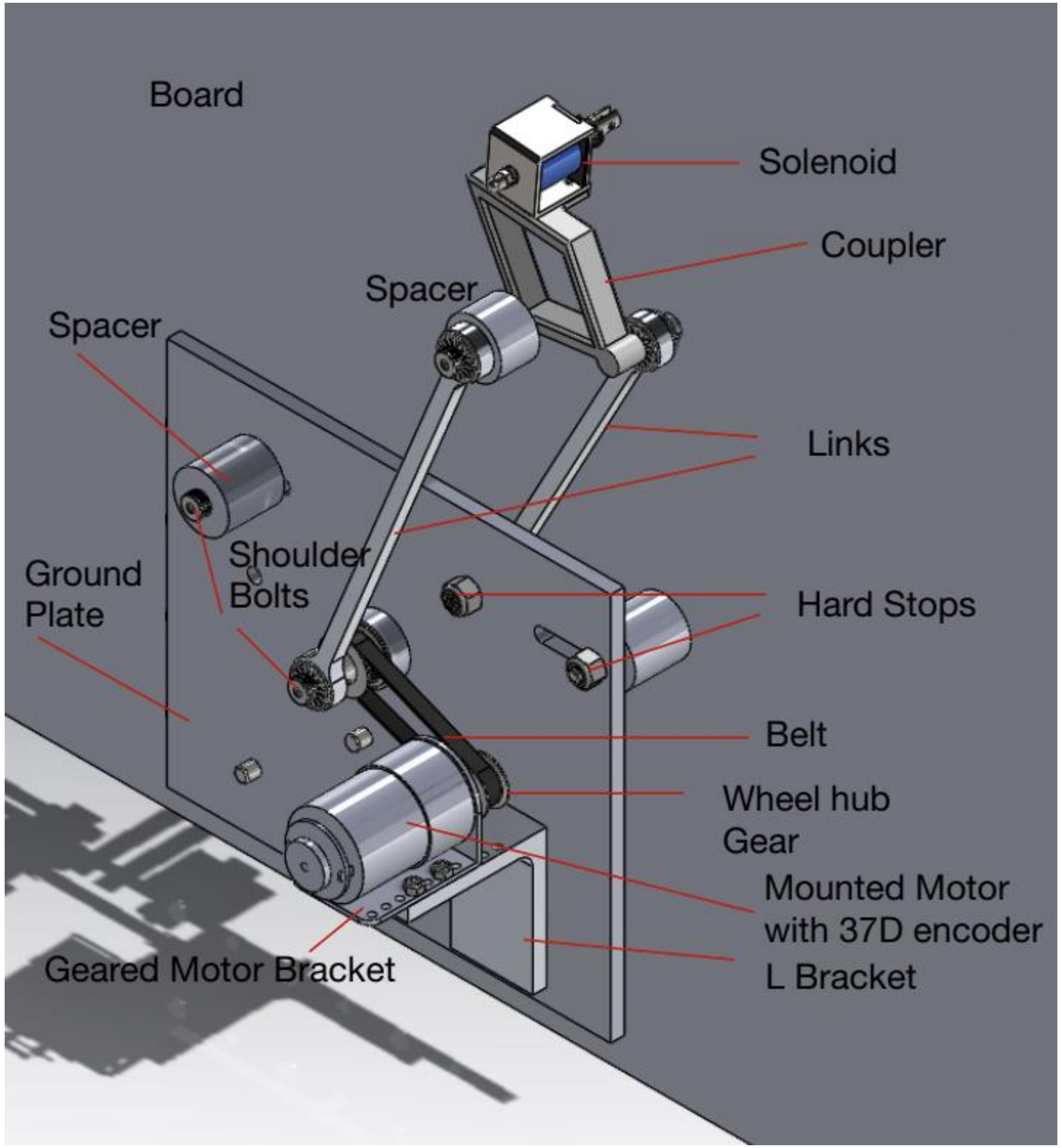


Each team member individually brainstorming linkage angles with select anchor points in the given area and then discussed as a team to decide which was the most ideal. Final design uses the three positions displayed.

CAD assembly



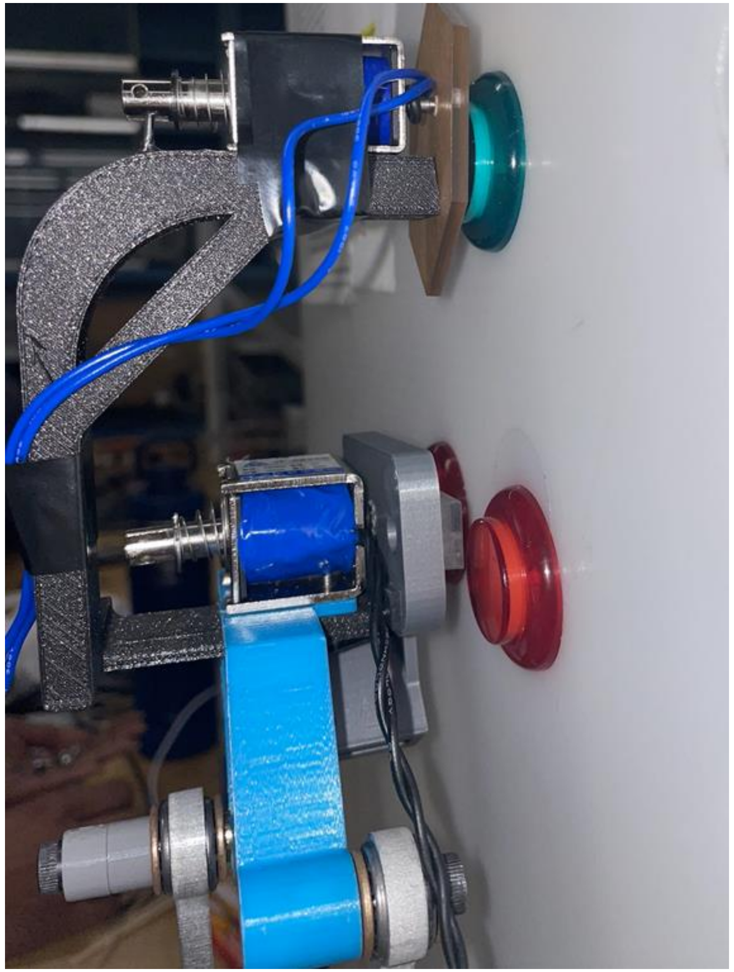




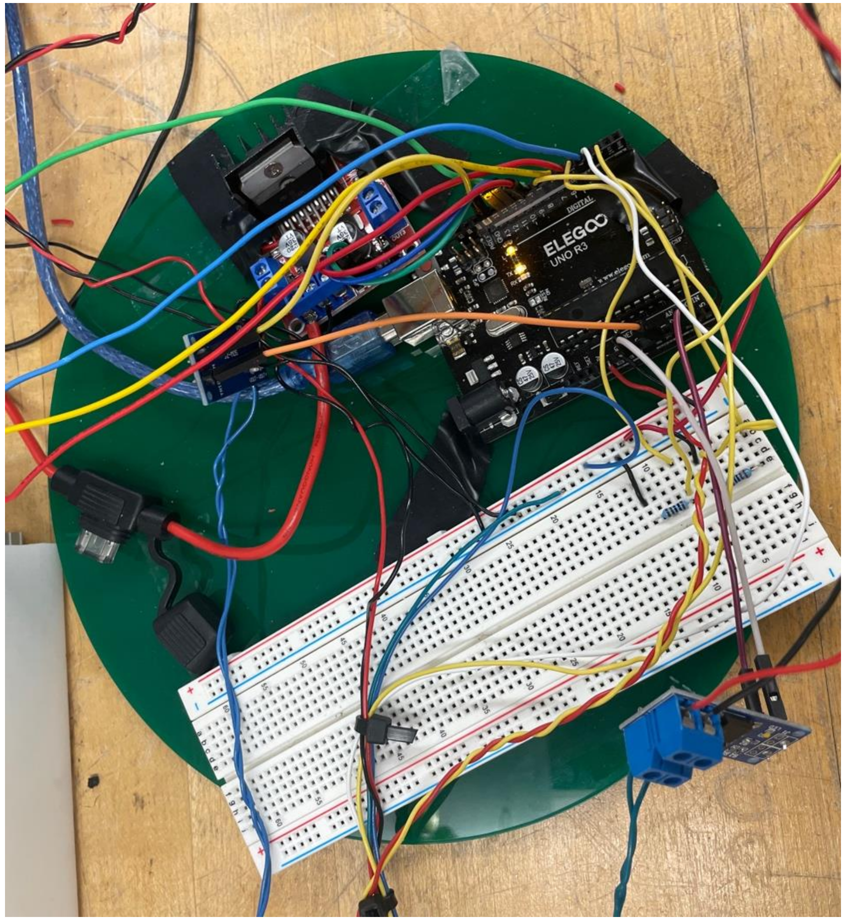
Fabrication and Assembly

Machined link with gear assembled

3D printed coupler and bonus button press mechanism



Electronics



Troubleshooting

I identified a major mechanical problem during my initial testing

Problem: Transmission ratio too low, not enough torque to consistently move our linkage system.

Ideal Solution: new gear to increase our transmission ratio BUT unable to purchase new/alternative parts

Initial Solution: increase belt tension as shown, but belt continued to slip

Action plan:

- Teammates looking for different approaches to maintain belt tension (e.g. alternating assembly, research modifying belt, etc.)
- I worked on developing and refining Arduino code so if problem is solved, we wouldn't fall too behind in our schedule

Testing → Success!

The odds seemed against us as it was still nonfunctional the night before demo.

Refusing to give up I raked through the lab's miscellaneous parts and found a compatible gear!

Modified by drilling out the press-fit pin (shown on the left)

Successfully achieved full motion to the linkage and reliable performance!

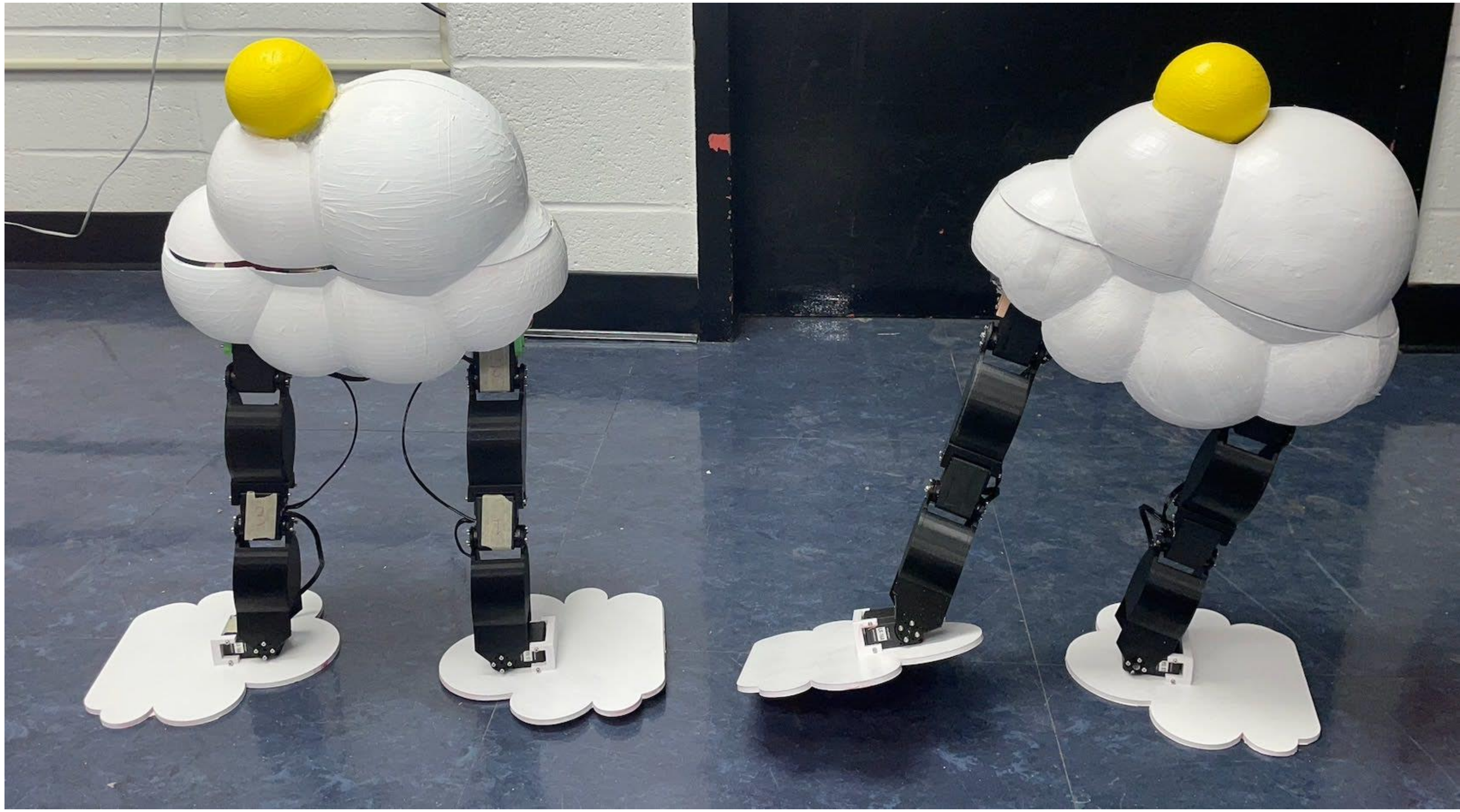
On the right – screenshots of assembly at button positions as they light up

Video: https://www.youtube.com/shorts/t_6y0Fk8vc0?feature=share

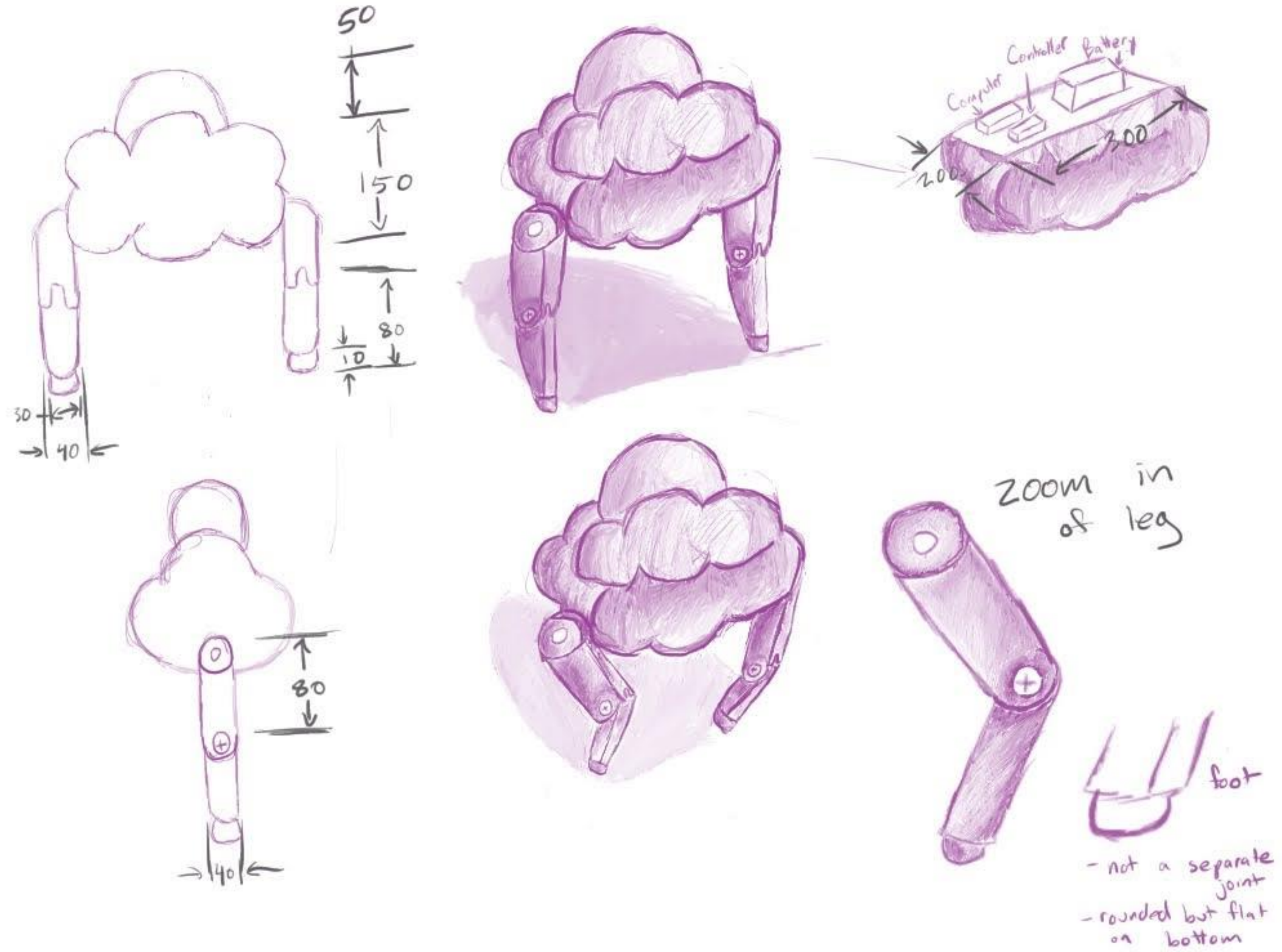
CloudBot – Bipedal Robot

Objective — Design a bipedal walking robot using provided kit and up to 3 additional motors of the same kind. A Raspberry Pi 4, DC-DC Converter, battery pack, and necessary cables were all also provided.

Skills: Freehand sketching, CAD (SolidWorks), Electrical Integration, Raspberry Pi, Python, Prototyping, Rapid Iteration

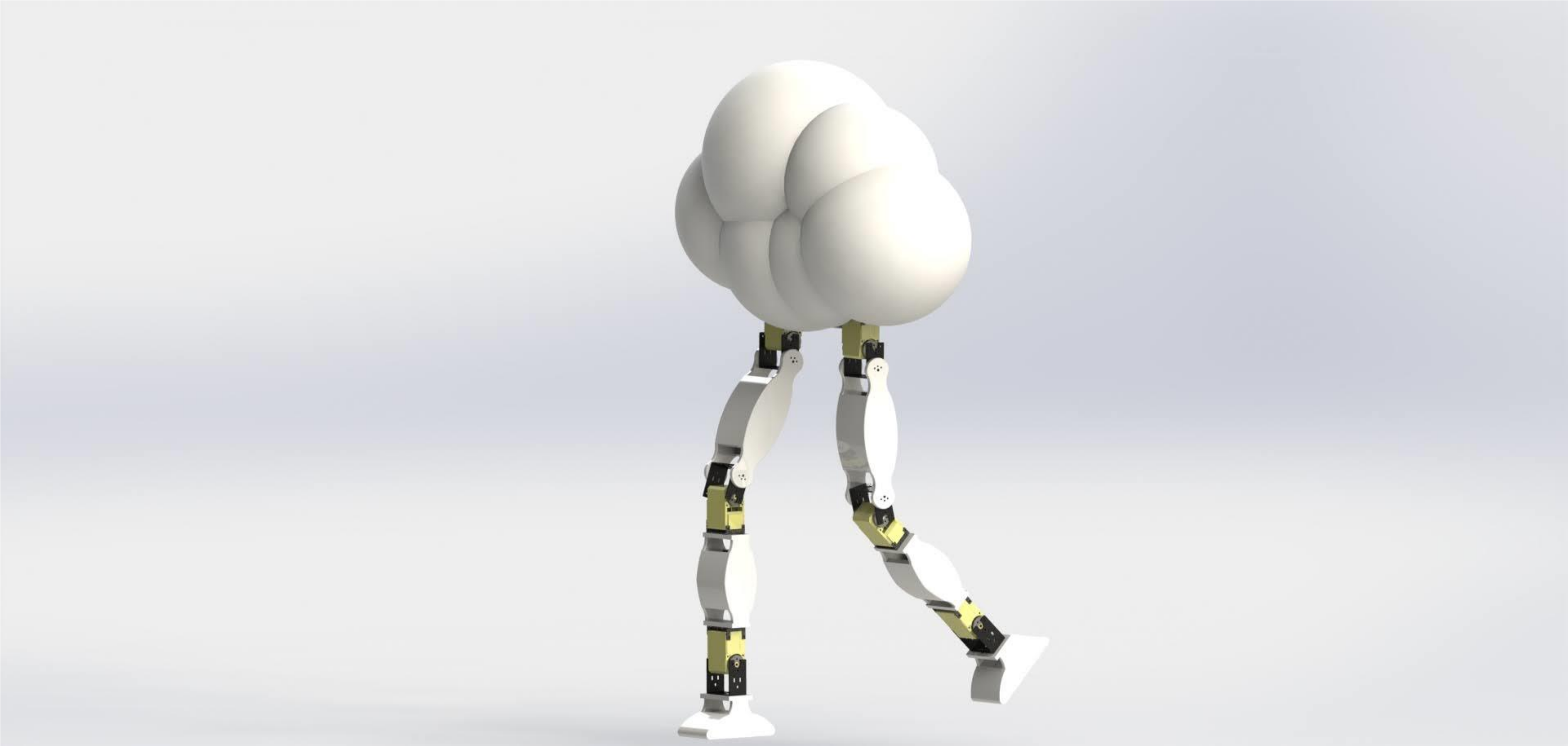


Concept Sketch

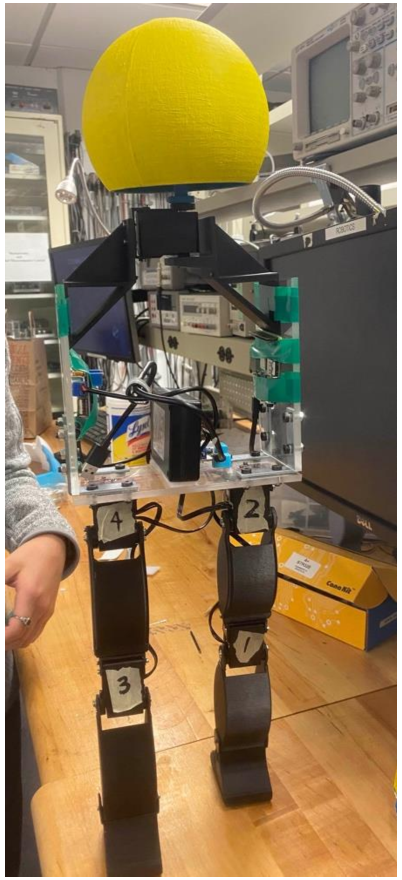


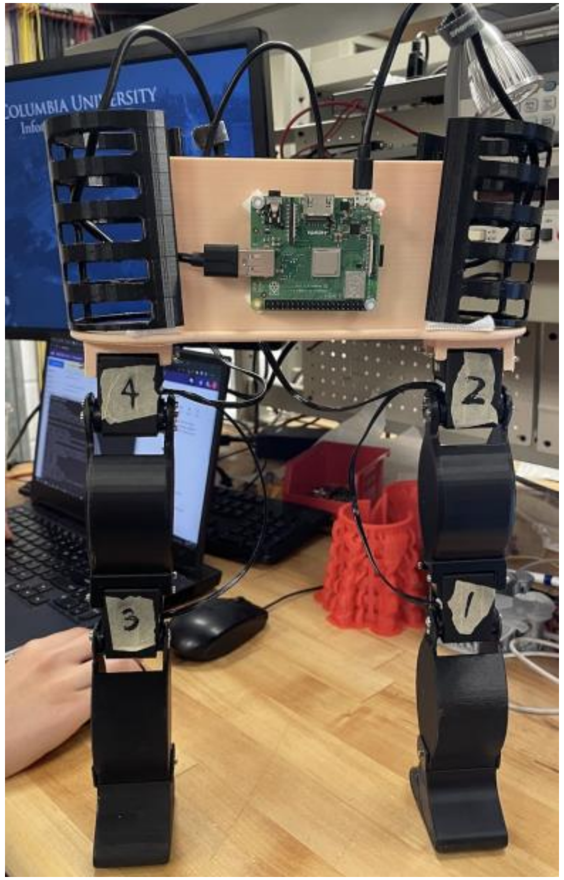
Design inspired by my appreciation for clouds.

Initial CAD

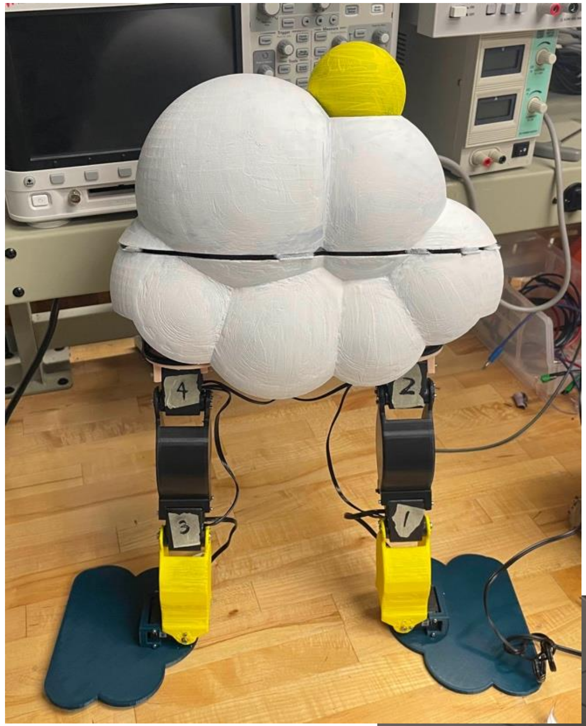


Prototyping & Rapid Iteration

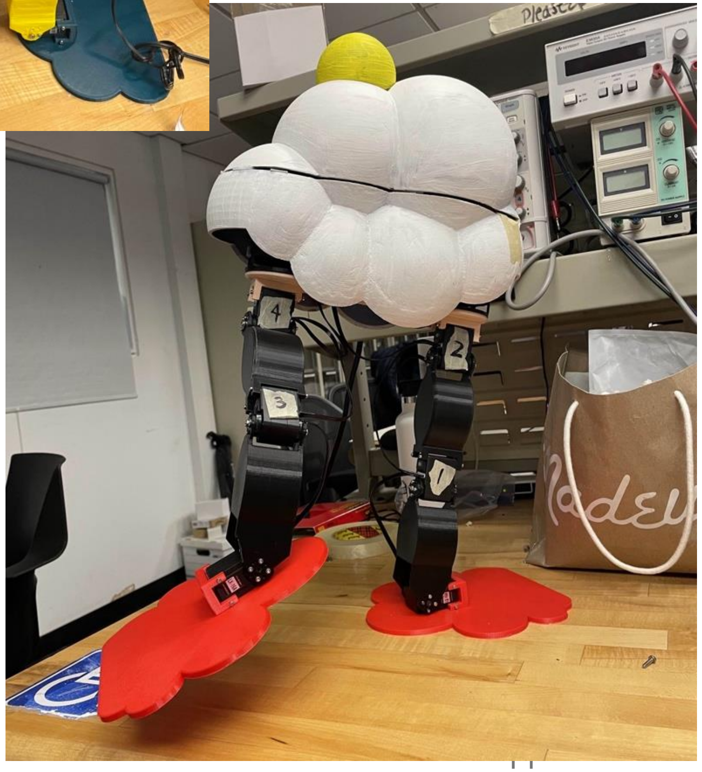




Chassis design improved for weight, assembly, and cabling

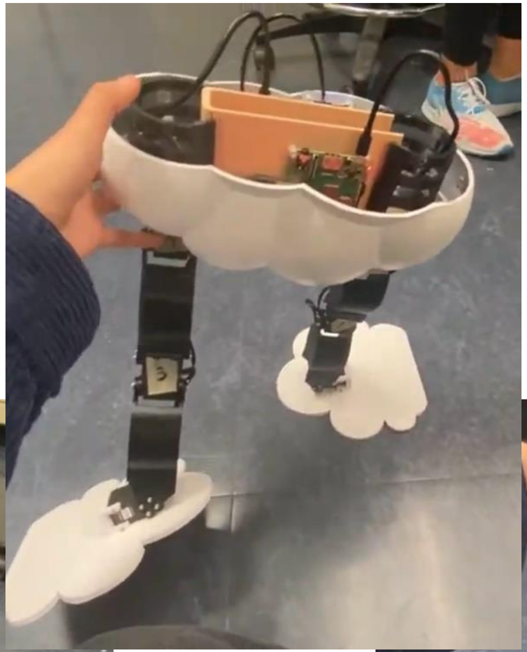


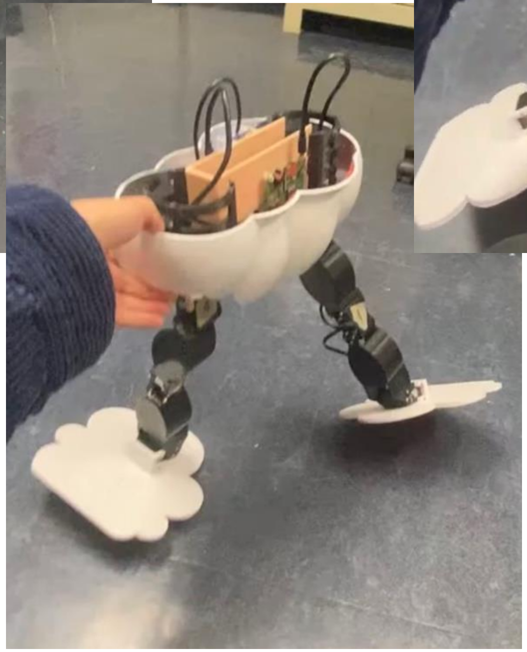
Bigger feet for better stability

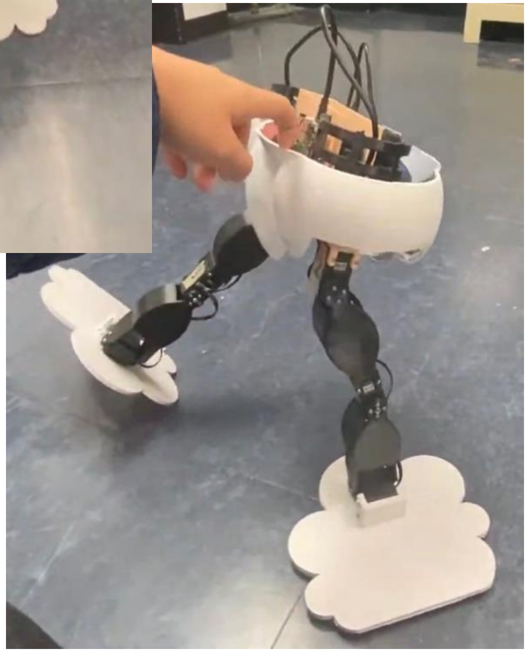


First 3D print of cloud body

Gait Testing & Optimization







- Final Walk: https://youtu.be/XSQKPm5_oK0
- Journey Video: https://youtu.be/GinokXPkORg

Redesigning Soccer Cleats to Reduce ACL Tears in Female Athletes

Project Website: https://nsm2154.wixsite.com/team-16

Requirements
- Solve a well-articulated need
- significant physical and/or simulation component related to mechanical engineering design.
- The prototype and/or simulation must have a mechanical function
- Solution has to fit problem. Cannot force solution onto problem.
- Reasonable scope, neither too small or too big (instructor approval)
- **Novel** solution

Chosen Problem: Female soccer players are up to eight times more likely to tear their ACL in comparison to their male counter parts.
Proposed Solution: Alternative cleat design with varied densities throughout outsole to achieve optimal valgus and knee flexion angle, reducing risk of ACL injury.

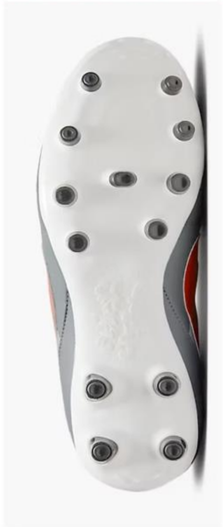
My **Primary** role: Experimental Lead (procedure, setup, data processing and analysis)

Skills: Experimental Design & Testing, CAD (SolidWorks), FEA, 3D Printing, Prototyping, Biomechanics, Data Processing & Analysis

Modeling and FEA Simulation

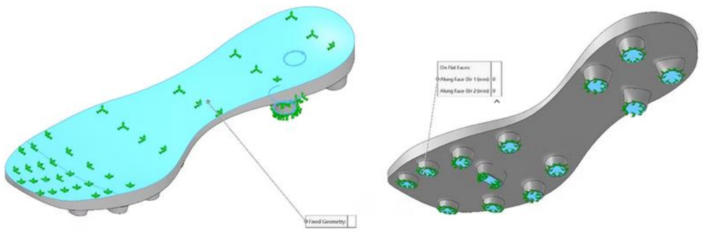
Our Model

Nike Premier 3 FG

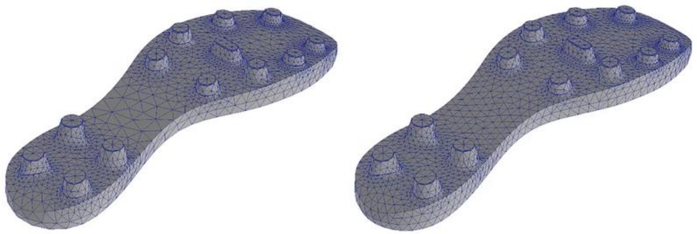


Cleat design itself was controlled so it was modeled based off the Nike Premier 3 FG, a popular and reputable cleat.

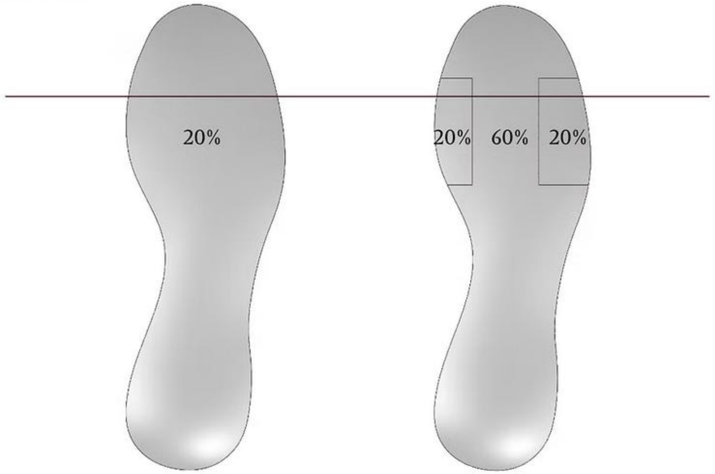
Supports



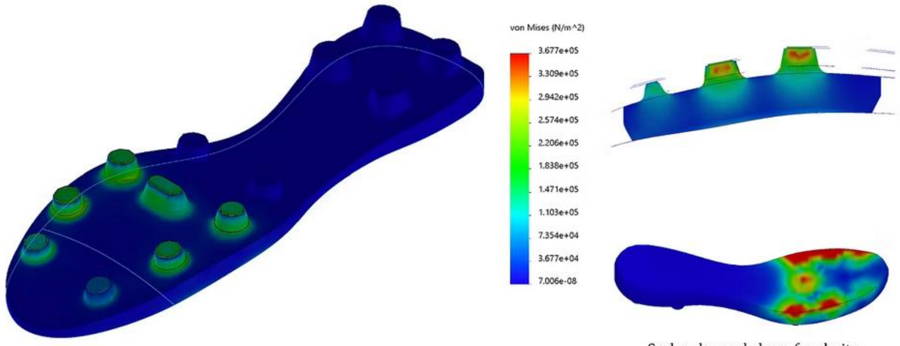
Setup of Mesh



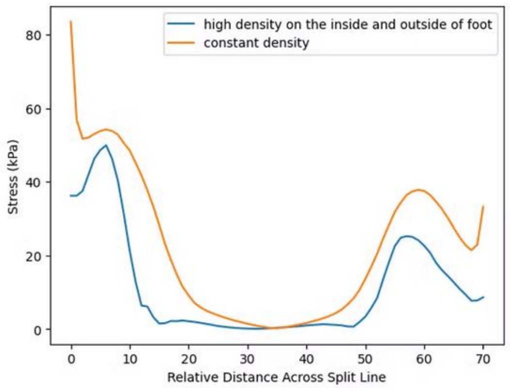
Stress Comparison

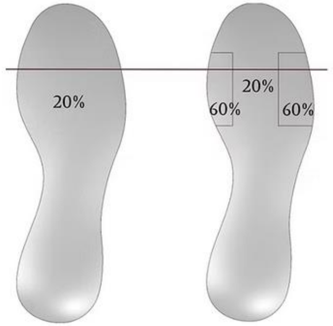


Stress Results



Scales changed above for clarity





constant density

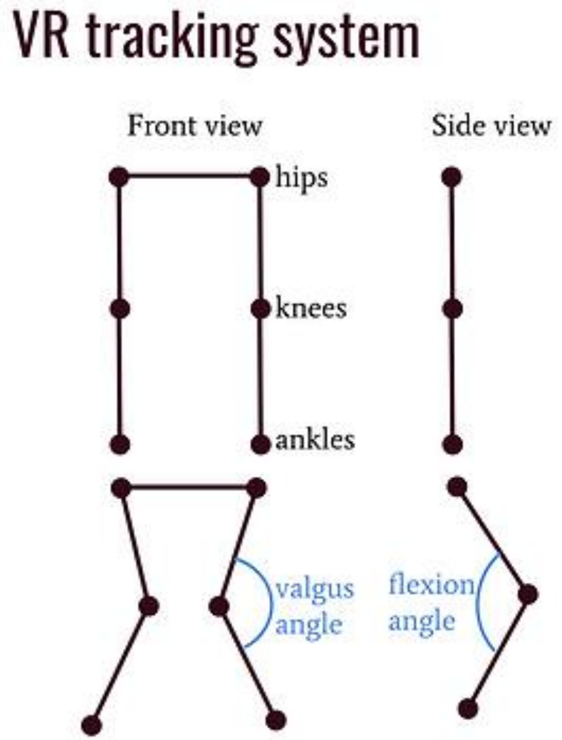
high density on inside and outside

Prototyping

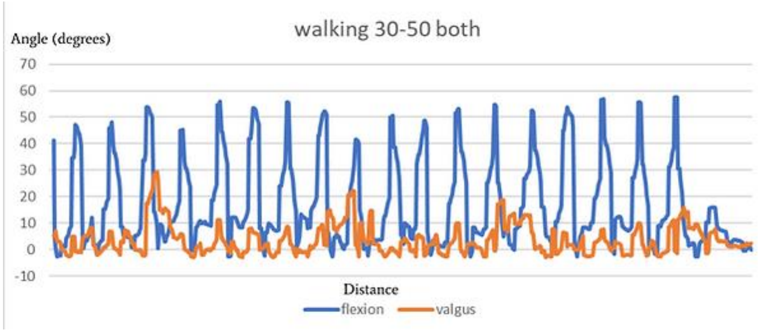
Experiment

Testing/Dependent Variable: Outsole density – printed seven different density patterns to be tested

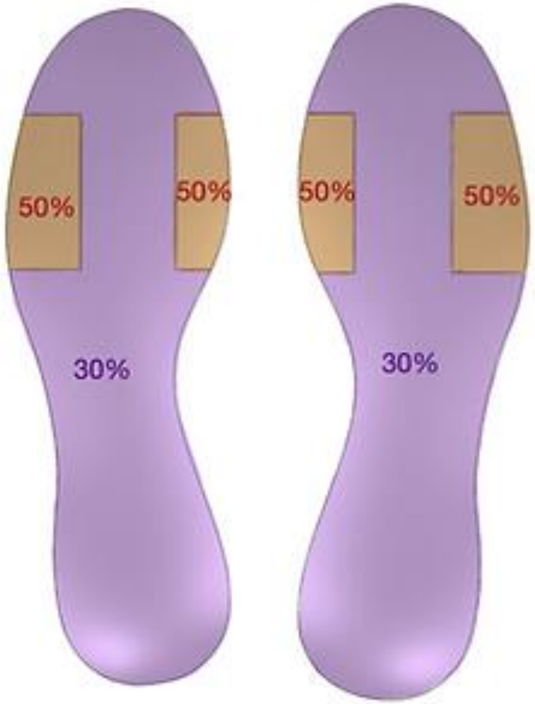
Procedure: Using a total of 4 VR trackers (one on each hip, one above knee, one above ankle) the flexion and valgus angles were measured as shown below. 5 athletic young adult females were tested while walking, running, and jumping.

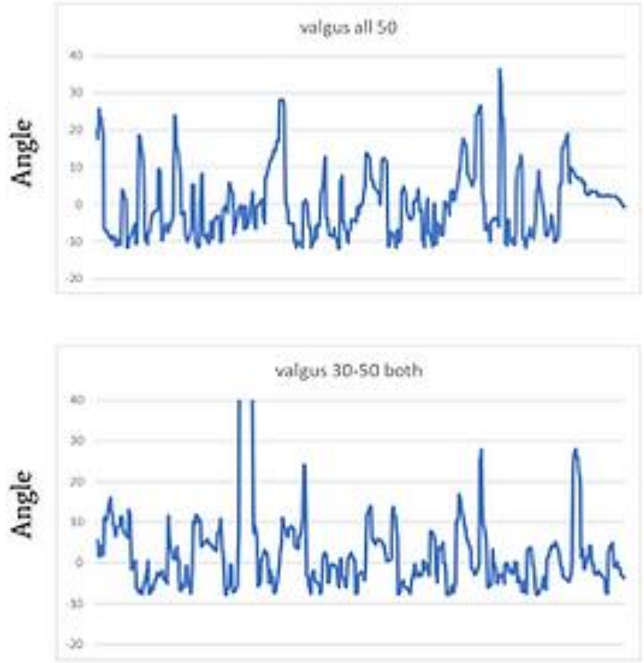


Results



Random trial raw data





Determined best pattern

Boxplot comparing patterns

Research Projects

Characterizing the Vibrational Response of a Fuel Injector

This research was a part of Boston University's Research in Science & Engineering (RISE) program (Summer 2018).

Introduction

Fuel injectors are electronically actuated mechanical devices which operate under high hydraulic pressures. The nozzle is designed to deliver fuel to an engine as a spray in a cylinder’s combustion chamber. Although vibration is typically avoided due to mechanical stress and noise, it can also be a useful diagnostic tool. In fuel injectors, cavitation collapse, fluid hammer, turbulent flow, acoustic waves, and needle actuation are all plausible causes for vibration. In this study, a laser vibrometer method is used to examine the cantilever vibration of a fuel injector to discern the void fraction of gas or vapor.

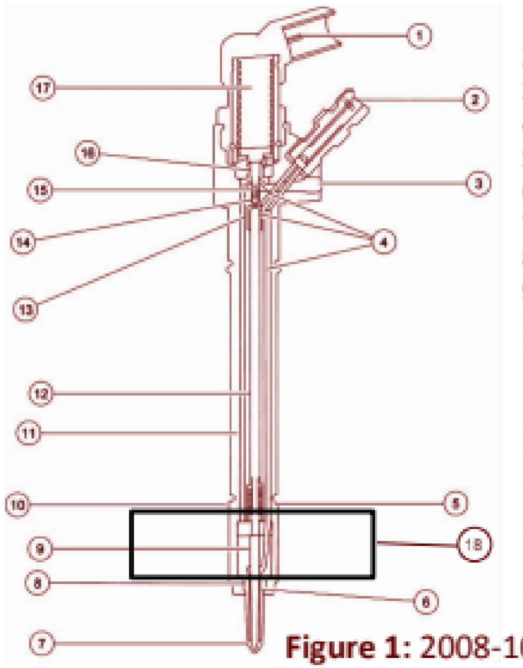


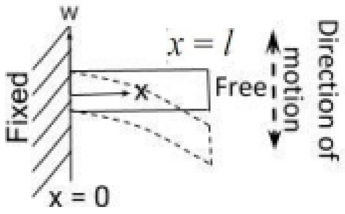
Figure 1: 2008-10 Ford 6.4L diesel fuel injector[2]

Theory

Flexural (Cantilever)

Equation 1a: Free vibration equation

$$\frac{\partial^2 w(x,t)}{\partial t^2} + c^2 \frac{\partial^4 w(x,t)}{\partial x^4} = 0 \, , \; c = \sqrt{\frac{EI}{\rho A}}$$



c = flexural wave speed

E = Young's Modulus (measures stiffness)

I = moment of inertia

ρ = density

A = cross-sectional area

$w(x=0) = 0$

$w(x=l) =$ free boundary conditions

Figure 2: Illustration displaying direction of cantilever oscillations

Longitudinal (Acoustic)

Equation 1b: Acoustic vibration equation

$$c^2 \frac{\partial^2 P}{\partial x^2} - \frac{\partial^2 P}{\partial t^2} = 0 \, , \; c = \sqrt{\frac{\beta}{\rho}}$$

c = speed of sound

ρ = density

P = pressure

β = bulk modulus

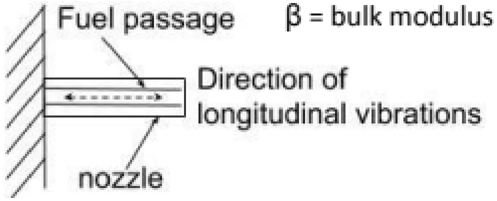


Figure 3: Illustration displaying direction of acoustic oscillations

Mode Frequency:

Equation 2: (a)Mode Frequency equation for flexural oscillations (b)Mode Frequency equation for longitudinal oscillations

$$\text{(a)} \quad f_n = \frac{\xi_n^2}{2\pi} \frac{1}{l^2} \sqrt{\frac{EI}{\rho A}}$$

$$\text{(b)} \quad f_n = \frac{nc}{2l}$$

n	1	2	3	4	5	6
Flexural Frequency(Hz)	2.366×10^4	1.483×10^5	4.152×10^5	8.136×10^5	1.345×10^6	2.009×10^6
Longitudinal Frequency of air(Hz)	1.056×10^4	2.112×10^4	3.168×10^4	4.224×10^4	5.280×10^4	6.336×10^4

Table 1: Calculated theoretical frequencies. Measured length as 16.24mm. Approximated inner radius as 2.84mm. Used Young's Modulus and density for stainless steel.

Mode Shape:

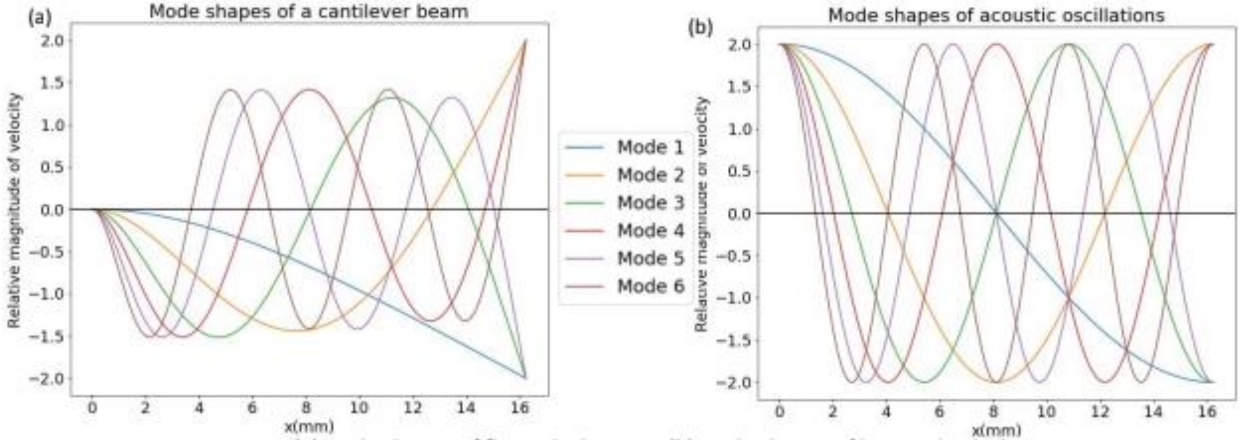
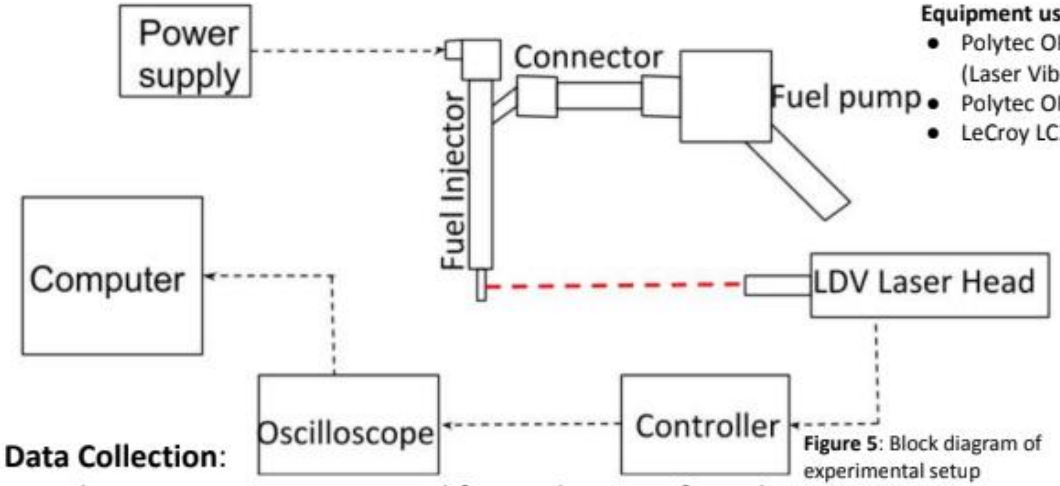


Figure 4: (a)Mode shapes of flexural vibrations (b)Mode shapes of longitudinal vibrations

Method



Equipment used:
- Polytec OFV 303 Sensor Head (Laser Vibrometer)
- Polytec OFV 3001 controller
- LeCroy LC334A oscilloscope

Figure 5: Block diagram of experimental setup

Data Collection:
1. Polytec OFV 303 Sensor Head focused on tip of nozzle.
2. Fuel injector actuated with high voltage (~90V) which excites the pintile.
3. The impulse response is measured by the LDV, the signal is then converted by the oscilloscope, and finally recorded by the computer.
4. Repeated along the length of the nozzle with 0.500mm steps

This procedure was first executed with the fuel injector empty, then it was filled with water.

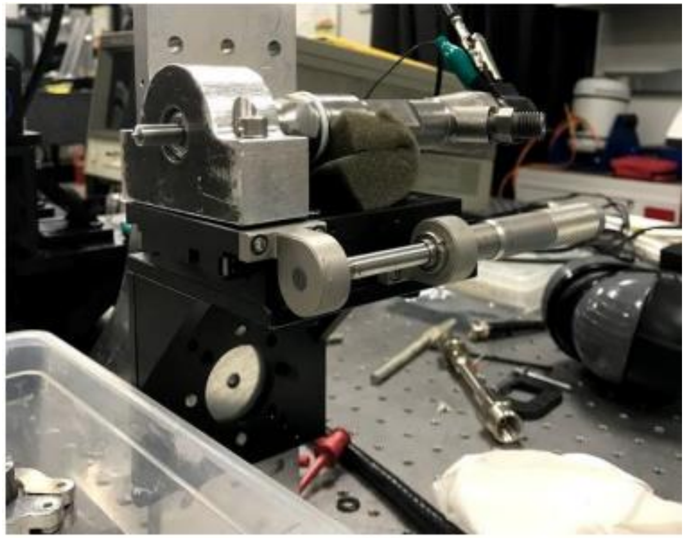


Figure 6: Mounted fuel injector

Figure 7: CAD part for fuel injector mount. The mount for the fuel injector was machined at BU Engineering Product Innovation Center.

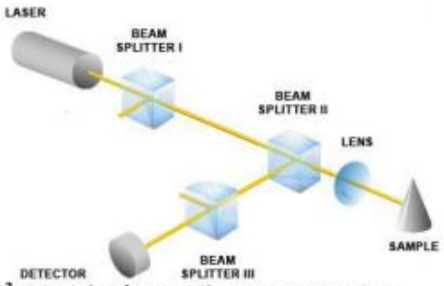


Figure 8[3]: Principle of Laser Vibrometer operation
The LDV system splits the beam of the laser into a reference beam and a measurement beam. The measurement beam is then reflected off the fuel injector and combined with the reference beam. The LDVs are capable of measuring both displacement and velocity, however, in this study, velocity is utilized because it is better suited for higher frequencies.[3]

The laser vibrometer utilizes interferometry and the Doppler effect to measure displacement and velocity on vibrating surfaces. The Doppler effect, the change in frequency of moving source observed from a stationary reference, is expressed with **Equation 3**[1]:

Equation 3: Doppler effect

$$f_{obs} = f_{source}\frac{a+v}{a}$$

Where:
f_{obs} = observed frequency
f_{source} = frequency of the source
a = speed of wave
v = speed of source

Results

Figure 9: Timeseries of velocity near tip of nozzle
(a)Data collected from empty fuel injector (b)Data collected from filled fuel injector

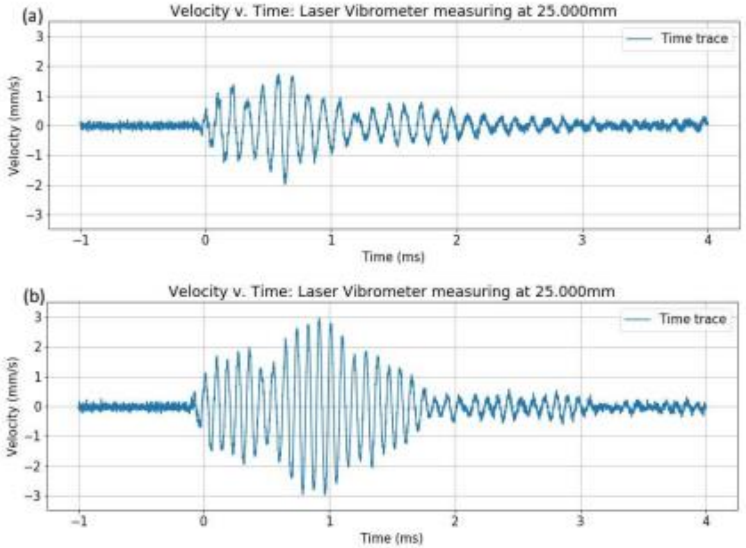
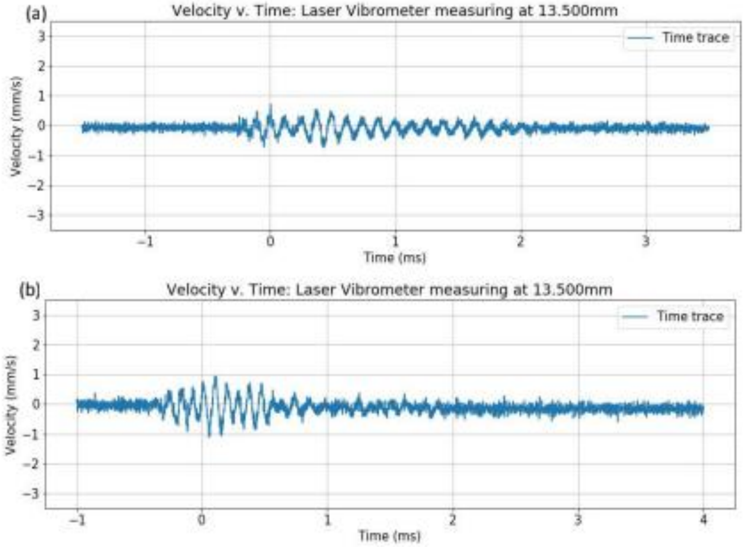


Figure 10: Timeseries of velocity near base of nozzle
(a) Data collected from empty fuel injector (b)Data collected from filled fuel injector



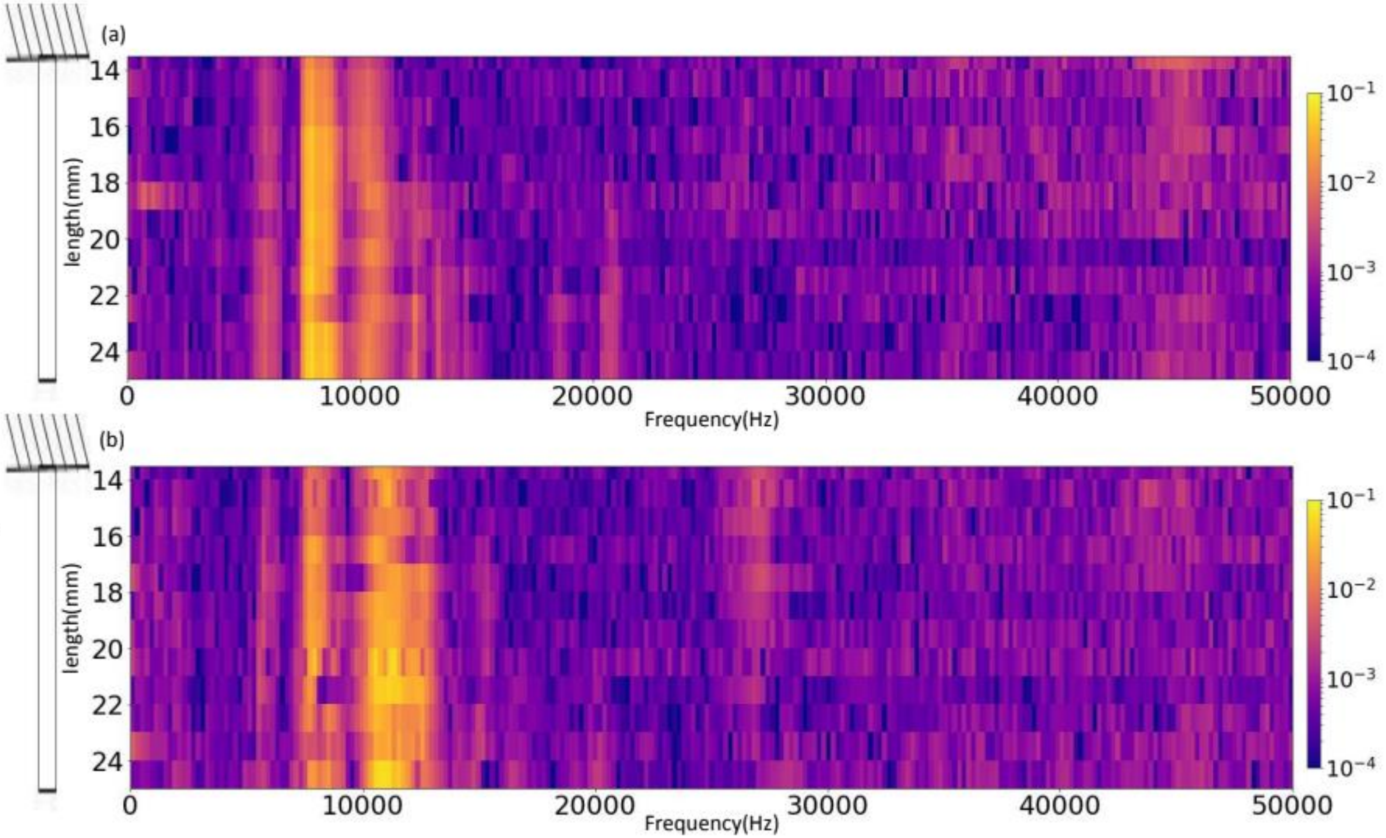


Figure 11: (a)Spectrogram for empty fuel injector (b)Spectrogram for full fuel injector

Conclusions

What frequency peaks are observed?
- Empty: 9000±200Hz
- Full: 11000±200Hz

Why does frequency increase as mass increases?
- In a simple spring-mass system, as mass increases, frequency decreases.
- In this study, the frequency increases as mass increases suggesting effects other than cantilever oscillations are occurring.

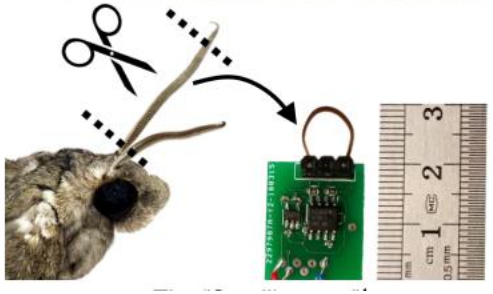
Which mode is occurring?
- The measured frequencies correlate more closely with the theoretical frequencies from longitudinal theory. However, longitudinal theory predicts maxima pressure at the base and tip and minimum pressure in the middle that are not observed. But according to the data, velocity is at its maximum at the tip but not at the base which correlates to cantilever oscillations.
- It is possible that cantilever vibrations are being driven by acoustic vibrations occurring in the air inside the fuel injector. However, the coupling isn't strong enough to display cantilever modes.

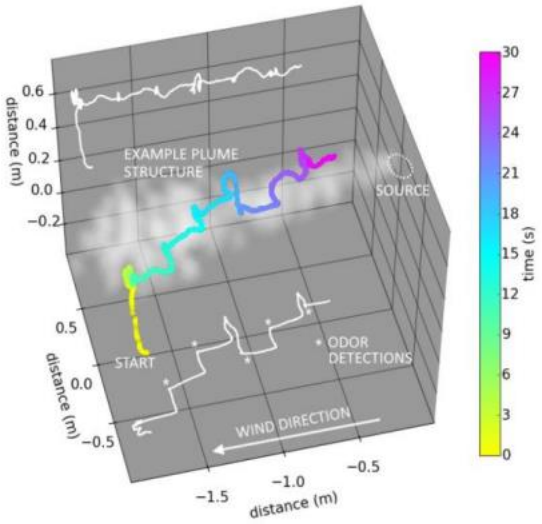
How to explore further:
- Excite the fuel injector in other methods — e.g. external impulse response or measuring driven frequency response
- Analyzing system with a computational model
- Flow fluid through the nozzle and study response length(mm) length(mm)

Design of Plasmonic Chemical Sensor

This research was conducted and funded by the Air Force Research Laboratory Scholars Program (Summer 2019).

Background





In order to detect chemical and biological agents, currently technologies require a person to use a handheld device to approach the unknown agent, putting the safety of the person at risk. In order to solve this problem, chemical sensors that can be attached onto any unmanned vehicle have been proposed.

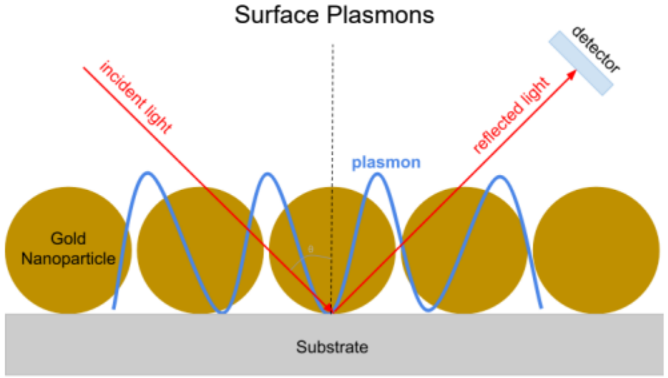
Notably, the “Smellicopter” designed in 2018, was able to detect and follow a moth’s chemical plume by using a a moth’s antenna as a chemical sensor. However, this approach had a short shelf life and was limited to only detecting specific moth hormones. Thus, additional research is being conducted in order to avoid these limitations.

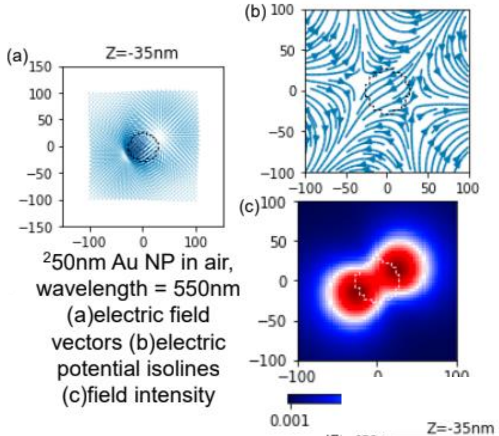
Model and Simulation

Surface plasmon resonances (SPRs) occur typically when light causes the free electrons in roughened (or nanoscale) noble metal surfaces to oscillate coherently. These oscillations give rise to a strong electric field which interacts very strongly with light at those frequencies. SPRs are very sensitive to changes in the surrounding environment.

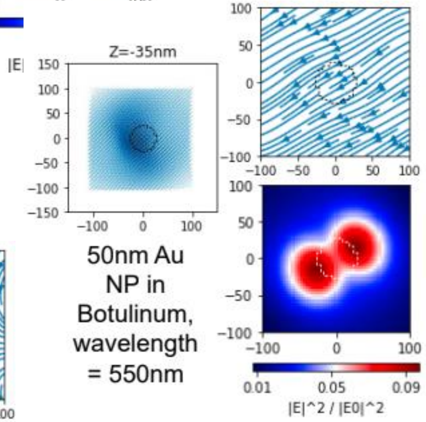
A single molecule in the vicinity of a SPR can cause an observable change in the SPR frequency. This arises from the change in dielectric environment. Although silver nanoparticles (NPs) have the sharpest SPR peak, it is highly prone to oxidation. Gold NPs are our most promising option due to their robust nature.

NPs with larger radii has been shown to be more sensitive to the environment than smaller ones. We aim to use the highly sensitive SPRs of metal NPs to detect a variety of chemical agents.

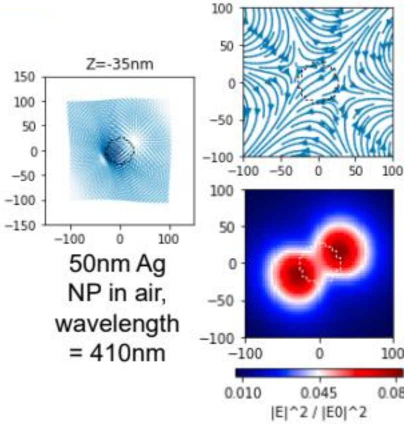




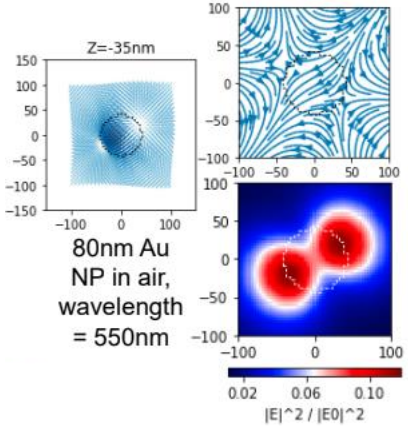
250nm Au NP in air, wavelength = 550nm (a)electric field vectors (b)electric potential isolines (c)field intensity



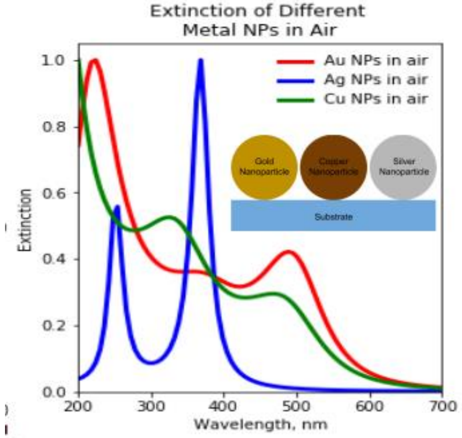
50nm Au NP in Botulinum, wavelength = 550nm

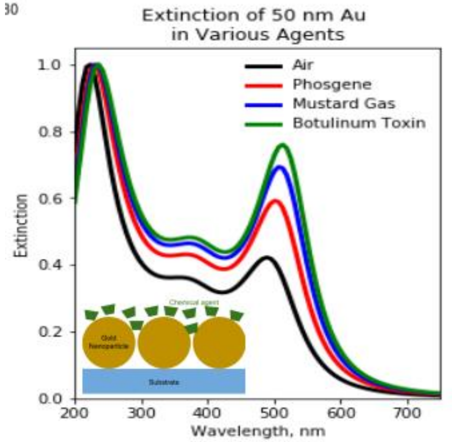


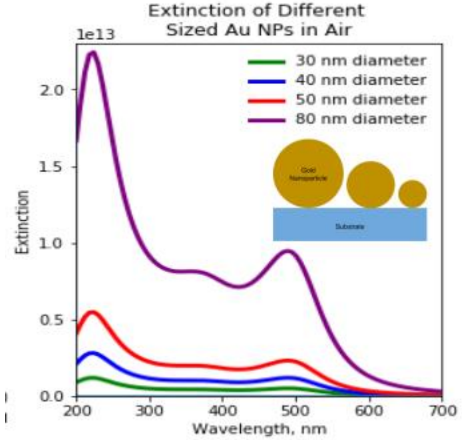
50nm Ag NP in air, wavelength = 410nm



80nm Au NP in air, wavelength = 550nm



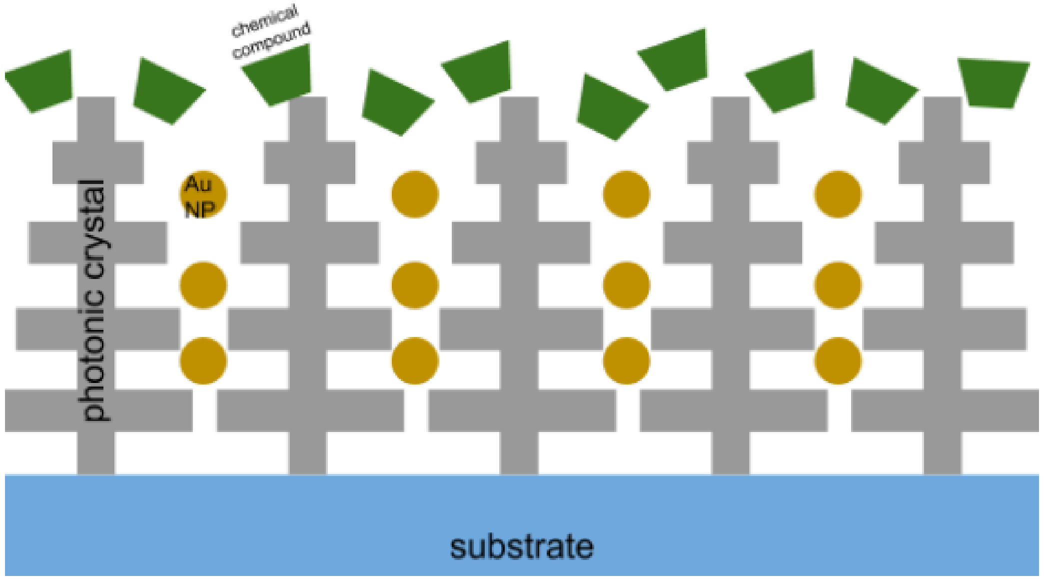




Future Direction

Photonic crystals found in the Morpho didius butterfly's wings have also shown a high sensitivity to the detection of chemical agents. We aim to use both of these phenomena to generate a hybrid chemical/biological sensor.

The sensor will first be modeled in COMSOL and simulations will be run using Python to optimize the design prior to fabrication of the device.



Proposed Model of Sensor